

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0013

Lammers 138 kV
May 04, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0013. The Interconnection Customer proposes a 261.8 MW solar photovoltaic facility interconnecting at the Lammers 138 kV substation in the Maryland area. The request was received on February 21, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 10 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,822,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

SUNSET to SADDLEBROOKE RANCH 138.0 kV: 103.9% loading (345 MVA) following the outage of the VIENNA OPERATIONS to EASTON 2 345.0 kV; DFAX = 12.7%.

For the loss of VIENNA OPERATIONS to EASTON 2 345.0 kV, the SADDLEBROOKE RANCH to LAMMERS SUBSTATION 138.0 kV loads to 103.3% of its 532 MVA rating. Project DFAX: 29.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SUNSET to SADDLEBROOKE RANCH 138.0 kV	Reconductor line	\$2,793,000	100% (\$2,793,000)
SADDLEBROOKE RANCH to LAMMERS SUBSTATION 138.0 kV	Reconductor line	\$5,029,000	100% (\$5,029,000)
	Total allocated to project		\$7,822,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.